
UNIT 1 INTRODUCTION TO INTERNET

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1.0 INTRODUCTION

Today it is the era of computers, to be more specific, the digital data. Every organization, small or big in size, doing business from a single site or multiplesite, from one nation or over the seas, not only uses computers to handle various transactions but also communicate information among various sites located over various geographically distant places. The means by which the information is exchanged among computers separated apart is known as the computer Internet. a network is; entities connected to each other to exchange some information. when entities are replaced by computing devices, it is the computer network. Digital information is made available in just one click from one end of the world to another end within fraction of seconds. The productivity of every organization irrespective of their field has been increased enormously by sharing of information in digital form through computer network. A Computer network is two or more standalone computers connected to each other to share information, resources, messages etc irrespective of their installed geographical location.

Further, in section 1.1 the objectives of this unit are listed, one should be able to achieve these after completion of this unit. In section 1.2 briefly discussed about the Internet. Network taxonomy, public and private networks and how Internet can be accessed are discussed in brief in this section. Section 1.3 discusses about the various network topologies. Network Models are discussed in the section 1.4. The two network models namely: OSI model and TCP/IP model are discussed here. Section 1.5 summerizes the unit. In section 1.6 solution and their answers are covered for the unit.

1.1 OBJECTIVES

After completing this unit, one should be able to:

- understand the term computer networks;
- differentiate between different types of computer networks;
- identify the advantages of using computer networks.
- identify the applications of computer networks;
- understand various network topologies;
- know the significance of network protocols;
- understand the two computer network models: OSI and TCP/IP reference model;

1.2 WHAT IS THE INTERNET?

The Internet is the biggest computer network with a world-wide scope. It is the network of networks. The Internet comprises lakhs of organizational networks of types: domestic or international, academic or business, and government, which altogether exchange various kinds of data/information. In routine we generally use the term 'net' for Internet. It connects billions of people through computers or mobiles from every corner of the world to share information. World's first general purpose computer network was ARPANET. ARPANET. The Bob Taylor is known as the father of the Internet. He initiated the project in 1966 to enable access to remote computers. As a result, in 1969 first time in a computer is accessed remotely. This was the start of what we see today around us in digital information and computer world.

The public Internet is a world-wide computer network i.e. millions of computing devices are connected to each other spread throughout the world. Computing devices may be desktop computers, laptops, mobile phone, Web TVs, installed with various kind of Operating Systems i.e. windows based, android based, linux/unix based mac-os based etc. In the Internet jargon, these devices are called hosts or end systems. End systems use network applications such as WWW (browsers) and e-mail etc to connect with Internet.

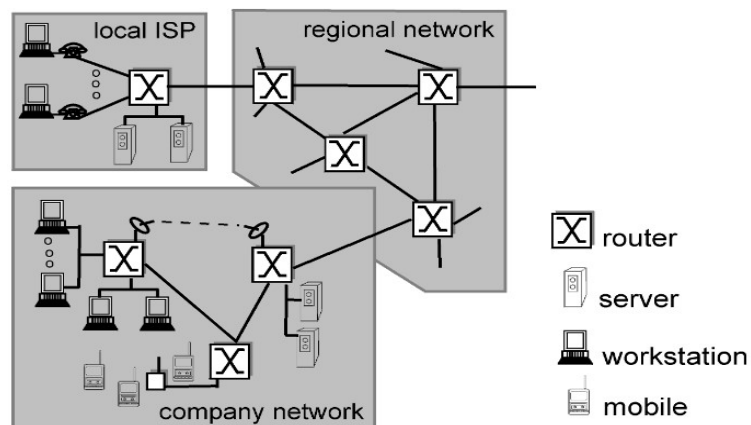


Figure1 : Overview of the Internet

1.2.1 ISP and Internet Backbone

ISP

ISPs (Internet Service Providers) are responsible for transporting of traffic in the Internet. Internet is very vast in terms of traffic, number of end systems, number of intermediate devices, interconnecting links and distribution of these over geographical locations. Hence huge number of ISPs manage the Internet. To efficiently manage and control the services of Internet, these ISPs are classified into three categories namely: tier -1 ISPs, tier -2 ISPs and tier -3 ISPs and this hierarchical model is known as 3-tier model. An ISP is categorized as tier -1, 2 or 3 based on the type of Internet services they provide.

Any one of us can become an access ISP as soon as we have an Internet connection. All we need to do is buy the required hardware/software (ex. router, modem, firewall etc) to allow other users/customers to connect to us. Adding new tiers and branches to the Internet is as simple as adding a new piece of Lego to an existing Lego construction.

Internet Backbone

Internet backbone is named for core network, can handle traffic at very high speed, interconnecting various networks to allow transmission of data over the worldwide network. Internet backbone mainly consists of tier-1 ISP's networks altogether. It connects networks of all tier -1 ISP and carries the majority of Internet traffic. Internet backbone mainly contains a large number of routers, switches and other intermediate devices interconnected with very high speed links (optical fiber links). Many companies may build their own backbone infrastructure for interconnectivity among offices at different geographical locations. Similar is the case with Internet backbones, where the computer connects to an access network of a tier -3 ISP, which further connects to the main Internet backbone (tier -2 or tier -1 ISP's network) to access the worldwide network.

1.2.2 Interconnection of ISPs

End systems or hosts situated at the edge of the Internet are connected to the rest of the Internet through this 3-tier model of ISPs, as shown in Figure. The network through which end systems connect to the Internet is called as the access network (i.e. local LAN or private network). Access network connects to the Internet via tier -3 ISPs also known as access ISPs and are at the bottom of 3-tier model. Access ISPs are the largest number of ISPs in the Internet. Tier -1 ISPs are relatively less in numbers and are at the top of the hierarchy. All three tiers of ISPs are same in many ways as any network, like they all have connecting links and intermediate devices i.e. routers and are further connected to other networks/ISPs.

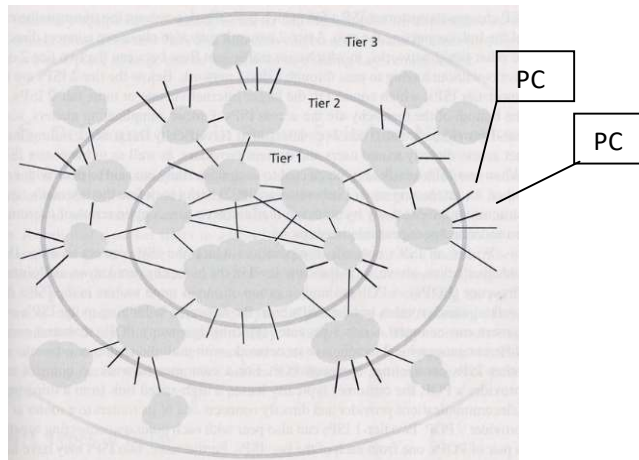


Figure 2 : tier model of ISPs

On the other hand, each of the tiers of ISPs are special and responsible to perform specific jobs. Tier-1 ISPs have links of speed of the range 622 Mbps to 2.5 to 10 Gbps. The routers of tier-1 ISPs must be able to handle traffic in accordance with extremely high rate links. Tier-1 ISPs are directly connected to each of the other tier-1 ISPs. The coverage of tier -1 ISPs is the entire Internet spread to all over the world.

Tier -1 ISPs are also known as Internet backbone networks. Some of the tier -1 ISP are Sprint, MCI (previously UUNet/WorldCom), AT&T, Level3 (which acquired Genuity), Qwest, and Cable & Wireless. Interestingly, no group officially sanctions tier-1 status; as the saying goes—if you have to ask if you are a member of a group, you're probably not. Tier -1 ISPs are connected to a large number of tier-2 ISPs.

Tier-2 ISPs have regional or national coverage, and are connected to only a few of the tier-1 ISPs. Thus, tier -2 ISPs need to route traffic through one of the tier -1 ISPs to reach a large portion of the global Internet. A tier-2 ISP is a customer of the tier-1 ISPs it is connected with, and tier -1 ISPs are the service providers for their connected customers.

Companies and institutions requiring high speed and less delay connect their network directly to a tier-1 or tier-2 ISPs. Service provider ISP charges its customer ISP a fee, depending on the transmission rate of the link connecting both. A tier-2 ISP network may choose to connect directly with other tier-2 ISP's networks. When two tier -2 ISPs are directly connected with each other, the traffic of these networks can flow directly between them without having to pass through a tier-1 network. Below the tier-2, are the tier-3 ISPs, which connect to the larger Internet via one or more tier-2 ISPs. At the lowest level of the hierarchy are the access ISPs. End systems or hosts are connected to Internet via access ISPs. When two ISPs are directly connected to each other, they are said to be peer ISPs.

The points (router) at which an ISP connect to other ISPs (of tier above, below or of same tier) are known as Points of Presence (POPs). A tier-1 provider has many POPs in its network spread over various geographical locations, where other ISP's routers and customer networks are connected. A customer leases a

high-speed link to connect its one of the router to a router in a provider's network POP. Furthermore, two ISPs may have multiple peering points connecting with each other at two or more pairs of POPs.

In summary, the topology of the Internet is complex, consisting of dozens of tier-1 and tier-2 ISPs and thousands of lower-tier ISPs. The ISPs are diverse in their coverage, with some spanning multiple continents and oceans, and others limited to narrow regions of the world. The lower-tier ISPs connect to the higher-tier ISPs. Users and content providers are customers of lower-tier ISPs, and lower-tier ISPs are customers of higher-tier ISPs.

1.2.2.1 Taxonomy of Network

Computer networks are generally classified according to their structure, physical scales, and transmission technology etc. According to the network's physical area, it can be classified as:

Personal Area Network (PAN): PAN is for a person. It is a network established for a person for a small area of the range of few meters. It can be wired (using USB data cable) or wireless (using infrared, ZigBee, Bluetooth). To establish PAN no third party device is required, the devices connect to each other autonomously and connect with each other. Examples of wireless PAN, or WPAN are: controlling TV with remote using infrared, connecting mobile headsets with mobile using Bluetooth, wireless keyboards and mice, wireless printers, wireless CCTV cameras etc using Wi-Fi, connecting printer, scanner, bar code scanners and game consoles using USB connection.

- **Local Area Network (LAN):** The physical scope of a Local area network is limited to a small geographical area, such as single building or a campus. Within an organization or institution personal computers are connected among themselves and establish a local area network to share resources i.e. printers, scanners etc and to share information.

- **Metropolitan Area Network (MAN):** The coverage of a MAN is spread over a city and limited to 100KMs range. MAN can be wired or wireless. TV cable network within a city and broadband network (wired as well as wireless) are examples of MAN.

- **Wide Area Network (WAN):** The coverage scope of a WAN is spread over a large geographical area and covers many cities, states and even countries. A WAN includes millions of computers connected located over a large geographical scale. The Internet is the largest Wide Area Network ever established.

There are three categories of wide area networks as follows:

Enterprise network: A network owned and managed by a single organization is named as enterprise network. It is the interconnected version of all the LANs of an organization/institution.

Global network: Networks of several organizations over a wide area collectively known as a global network.

Internet: A network of networks of broad area category. It is the largest network worldwide. It includes many LANs, WANs, MANs, and millions of computers. No single authority controls it; rather the local, national authority controls every segment of the Internet.

In addition to above, computer networks can also be classified as per the following characteristics:

- **Topology:** topology is how computers are arranged physically i.e. the physical arrangement of computer systems in a network. Widely known computer network topologies are: bus topology, star topology, ring topology, and mesh topology.

- **Architecture:** based on the architecture, networks can be classified as:

- peer-to-peer network or
- client - server architecture.

Transmission Technology: according to the transmission technology networks can be classified as:

- Broadcast networks.
- Point-to-point or Switched networks.

In a broadcast network a single communication channel is used which is shared by all the hosts connected to the network. In a broadcast network a message sent by any member reaches to all the members in the network. This mode of transmission is known as broadcasting. The sender sends message to all the computers connected to the network, the receivers capable of and are interested to receive the message may get it. To send a packet to all the nodes (not known) in a network, a special address: broadcast address is used as the destination address in the packet and it is called 'Broadcasting'. To send a packet to a group of nodes (all group members are known), 'Multicasting' is used. One possible way to achieve Multicasting is to reserve one bit to indicate multicasting and the remaining (n-1) address bits contain group number. Each machine can subscribe to any or all of the groups. Some of the common broadcast networks are the packet radio networks and satellite networks etc.

1.2.2.2 Standard Internet Protocols

Some rules are necessary to make any communication successful, this set of rules is known as the protocol. A protocol allows computers to talk with each other. A protocol defines the data type may be exchanged, set of commands used to exchange (send/ receive) data, and the way data to be transferred are how it is confirmed that data is received.

In human communication with spoken language also, some protocols are to be followed to communicate or share or exchange any information. Every spoken language defines set of rules to be followed while communicating. Like to make communication effective the first and basic rule to be followed in human communication is to use the language understood by both sender and the

receiver. A person from India can communicate with a person from any other country if both of them know (speaking and listening) a common language (i.e. Hindi or English or any other). In similar way, a hardware device can connect with other hardware supporting the same protocols irrespective of their manufacturer or type of device. Like an Email sent from an iPhone can be received on an Android device by the use of a common mail protocol used by both.

In the Internet various protocols are designed for networking applications. Like Ethernet protocol is used in wired network, IEEE 802.11 protocol is defined for wireless communication. An Internet protocol suite is a set of standard protocols used for transmitting data over the Internet.

Standard Internet protocols may be classified into four major categories:

1. **Application layer** – Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP).
2. **Transport layer** – Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
3. **Network layer** – Internet Protocol (IP) version 4 and version 6
4. **Data link layer** – Ethernet, IEEE 802.11 (Wi-Fi)

Application layer protocols are designed to facilitate Internet users to access various Internet resources like HTTP for accessing webpages, SMTP for sending emails etc. A number of application layer protocols are available to use. Transport layer defines the way data is transmitted, received and confirmation of receipt correctly. Transport layer provides two ways to transmit data: reliable data delivery and unreliable data delivery. TCP provides reliable data delivery meaning that it is confirmed that sent data is received correctly and intact whereas UDP provides unreliable data delivery meaning that data is sent and confirmation of receipt is not bothered. Network layer protocols are responsible to route the data in the Internet.

IP is responsible to provide the address of host and delivery of it to the destination host is the responsibility of the routing protocols. Protocols of data link layer are responsible for node to node data delivery. Data link layer protocols are dependent on the kind of topology used, like for wired network Ethernet protocol is used whereas for wireless topology IEEE 802.11 protocol is used. Apart from discussed here many other protocols also available in the Internet networking.

1.2.2.3 Public Network & Private Network (Intranet)

A public network is an open network i.e. anyone is allowed to connect to it. Internet is the best example of public network. Though both the public and private networks are there is no technical difference between a private and public network in terms of hardware and infrastructure, except for the security, addressing and authentication systems in place. In a public network no or very few restrictions or access control rules are applicable to get connected and access resources of the network. In private networks least security measures are applicable therefore, it is the least secure network and

one has to be very careful with respect to the security aspects while connecting to a public.

A private network is managed and controlled by an authority and access to a private network is restricted. Network of school, university or any organization are examples of private networks. In a private network the network administrator can apply rules and regulation on, how to access, what to access and how much to access the Internet. Private network is sometimes also known as Intranet. Intranet means within the network i.e. a computer network for sharing information, and computing services within an organization, with no access to outsiders.

In public networks public domain IP addresses are used while in a private network private domain IP addresses are to be assigned to the systems. In a private network there is a requirement of NAT (network address translator) which converts private IP address to public IP address before going into the public network and vice versa. Private domain IP addresses used in a private network or Intranet can be reused in other private network but, in a public network a public domain IP address can be used on a single system. While connecting to the public network i.e. Internet, one needs to obtain public IP address which could be obtained from your ISP.

1.2.2.4 Accessing the Internet

Access to the Internet can be available using either of the three methods namely: telephone network, cable network, and wireless network.

1.2.2.4.1 Telephone Network

The PSTN (public switched telephone network) is the telephone network used for voice-communication and spread throughout the world. For data installing a separate infrastructure in parallel to the existing PSTN is not easy and a worthy solution, instead the existing infrastructure for voice communication is used for the data communication also. Hence the telephone network system is inseparable/integral part of the computer data transmission. The very first data communication over telephone lines is implemented through dial-up setup. Further its enhanced version: dedicated lines came into existence.



Figure 3: PSTN Network

Dial-Up Lines

Setting up data connection over existing analog telephone lines require to install a modem on both the sides (sender side and receiver side). Establishing a data connection over the telephone line using **dial-up setup** is as similar as establishing a voice call over the same. Before sending the computer data, a connection is required to be established which is temporary in nature between sender and the receiver. To establish the connection modem on sender side dials the telephone number of the receiver side's modem. Once the connection is established data transmission can takes place. After the completion of the data transmission, either side of the communication can turn down the connection. During the communication no other connections can be established between end devices and even at a time either a data or a voice communication can be possible. The speed of the data transmission is very slow over the dial-up connection. The maximum speed could be availed is 56kbps over a dial-up connection.

Some of the advantages of using dial-up connection are: Low cost, and Availability

Some of the disadvantages of using the dial-up connection are: Low data transmission speed, Requires phone line, hence could not use it for both data and telephone voice communication, connection has to be reestablished each time, and Route busy: no other connection could be established between communicating end devices.

Dedicated Lines

Connecction established using dedicated lines does not require to dial the telephone number of the other end. A dedicated line establishes a separate connection for the data from the voice connection. It is always-on type of connection i.e. there is no need to reestablisheda connection each time it is used. Dedicated lines provide a high quality connection for data transmission than a dial-up line. Dedicated lines are classified into two categories namely: analoglines and digital lines. In present scenario the digital lines are widely used to connect end users to Internet worldwide. The data transmission over digital dedicated lines is much faster than that over the analog lines.

Some of the digital dedicated lines are ISDN lines, DSL.

Integrated Services Digital Network (ISDN)

Data transmission over Integrated Services Digital Network (ISDN) is much faster as compared to the dial-up lines. ISDN also uses standard telephone lines for data transfer and was one of the popular technique for data communication. An ISDN modem is required to be installed at both the communicating ends. The end used should not be out of the range of 3.5 miles from ISDN modem of the telephone connection providing company. Thus, ISDN is suitable to urban areas only and is not an option for users located in rural areas far from the cities (where ISDN modems of telephone service providers are installed). Data can be transferred at the rate of 64kbps over ISDN. Unlike dial-up line, speech and data/information can be transmitted over same line.

Digital Subscriber Line (DSL)



Figure 4: DSL access to Internet

DSL is used for transmitting information at fast speed using the existing telephone lines. The user can use both telephone and computer data transmission at the same time over the same line. A DSL modem is required to be installed at the end of the line to enable the computer to connect to Internet.

Unlike dial-up line there is no need to dial the telephone number of the other end to establish the connect, rather this type of connection is always-on connection.

Asymmetric digital subscriber line (ADSL) is a type of DSL widely used for Internet connection. One of the reason of popularity of ADSL is the higher downstream (downloading) rate than the upstream (uploading) rate. Most of the Internet users use the Internet services to download the contents from Internet making it more popular.

1.2.2.4.2 Cable Network

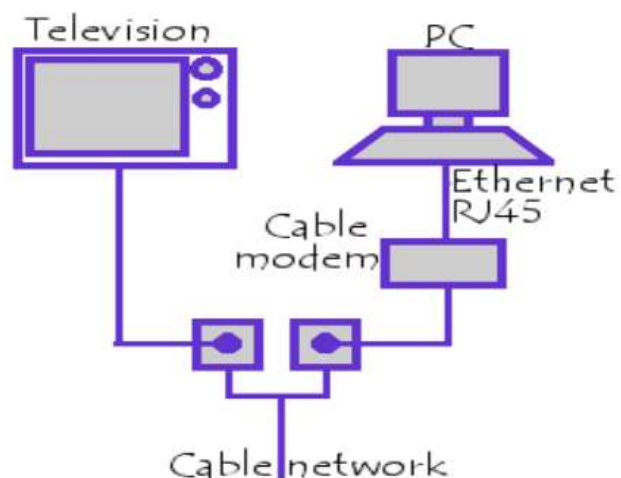


Figure 5: Cable Network

In a cable network cables are used as transmission medium in the network. Cable Internet uses the television cable infrastructure to provide the Internet accessibility. This is also called broadband Internet access. Cable modem device is used for Internet access in the cable network. Coaxial cable is used in the cable network for connectivity. Cable Internet provides faster Internet speed of upto 1 Gbps than dial-up and DSL connections.

Cable Internet access provides last mile access i.e. the cable network of television infrastructure is used from Internet service provider to end user. In today's scenario the cable Internet access is suppressed by fiber optical network, wireless network, and mobile networks.

Advantages of using cable Internet access: It provides high speed data transfer, it is convenient to use and easy installation, and it doesn't use telephone lines thus doesn't affect the telephone lines.

Disadvantages of using Internet access: it is costlier than the dialup and DSL connection, it is less secure as compared to the dialup or DSL connection, This service is not available with all cable Television networks.

1.2.2.4.3 Wireless Network

Wireless networks is much cheaper to install and maintain. Wireless network infrastructure does not require setup of wired network. A wireless modem/router/access point is required to establish wireless network. Many wireless technologies are available to establish wireless network. Bluetooth, InfraRed, Wi-Fi etc are some of the wireless technologies commonly used to establish a wireless network. Wi-Fi (wireless Fidelity) is one of the most widely used technology for wireless networks. A Wi-Fi enable modem has a typical range of approx 90 mtrs. The connection speed may reach up to 1544 Kbps.

1.2.2.5 Internet Services

Using Internet services, one is able to access information available on the Internet. These Internet services can be classified as four major categories: Communication services, information retrieval services, web services, and world wide web.

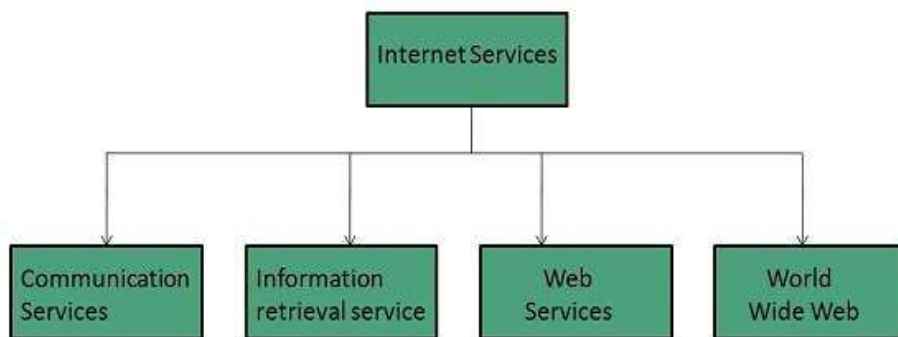


Figure 6 : Internwet Services

Communication Services

Communication services are used to exchange information over the Internet. One of the commonly used communication service is the Electronic mail. Electronic mail service is used to exchange the messages in the form of electronic messages. Other communication services are: Telnet – used to establish and exchange messages between remote systems, Internet Relay Chat (IRC) – it is used to communicate in real time, Internet Telephony (Voice over IP also known as VoIP): the internet users are allowed to talk using Internet services by making a call, Instant Messaging (IM): IMs allows real time chat between users and group. Eg. gtalk (google talk), Yahoo messenger etc.

Information Retrieval Services

Information retrieval services are used to download or extract information from the Internet. One of the commonly used information retrieval services in Internet is FTP (File Transfer Protocol): it enables the Internet users to transfer files to or from the Internet.

Web Services

Web applications use web services to interact with each other.

World Wide Web

World wide web (WWW) provides a way to retrieve documents located on servers located on different geographical locations over the internet. WWW documents may include audio, text, visuals, graphics and links to other documents (hyperlinks). Hyperlinks are used to navigate among the documents.

Check your progress 1

Q.1 What is Internet ? Explain.

Q.2 What is Internet Protocol? What internet protocol is classified?

Q.3 Define DSL and ADSL?

1.3 NETWORK TOPOLOGY

Topology of a network is defined as: the physical and logical arrangement of the nodes in the network. How nodes are interconnected defines the topology of the network. Topology could be defined for a wired network as devices are wirelessly connected and free to move and change their physical locations. Topology of the network directly affects the performance of the system. Before deploying a network it is very important to select appropriate topology to be installed. Topology of the network not only affects the performance of the network but also the installation cost of the network. The parameters to be considered in selection of the topology are: the size of the network, number of nodes in a network, the purpose of network, and capacity of fault tolerance etc. In computer networking network topologies are classified into four major categories namely: Bus topology, Ring Topology, Star topology and Mesh topology.

Bus Topology

In a bus topology nodes /computers are connected to a common link (also known as backbone link). Sometimes bus topology is also called a line or backbone topology. The data flows in single direction on the common channel. In a bus topology at a time only one computer can transmit the data. If more than one computer transmits data at a time, a collision occurs. One of the disadvantages of bus topology is single point of failure i.e. if the backbone link or the common link gets faulty the overall topology gets down.

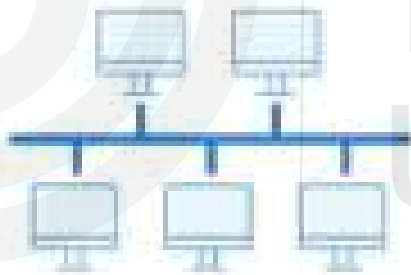


Figure 7: Bus Topology

Some of the advantages of bus topology :

- Installation wise bus topology is cheaper as compared to others.
- Easy to install
-

Some of the disadvantages of bus topology:

- If the backbone link is in fault, the communication of whole topology fails.
- At a time only one computer can transmit the data, if this rule is not followed collisions occur.
- Any information available on the common link is accessible to all connected computers, leading to unsecure communication.

Star Topology

All the computers/devices are connected to a central switch/hub/server forming a star like topology. All the communication in star topology take place through the central node. The centralized node can act as a simple forwarding device or can also act as an intelligent firewall / filtering device.



Figure 8: Star Topology

Some of the advantages of star topology :

- Easy to install and extend.
- Robust toward fault tolerance.
-

Some of the disadvantages of star topology:

- Single point of failure, i.e. if the central node fails, overall system crash.
- Performance of the topology is depended on the central node's performance.

Ring Topology:

Each device/computer is connected with two devices forming a ring type of tructure. In general the data can move in one direction in the ring, but by installing another cable the data can move in reverse direction also. A token message keeps on rotating in the ring topology.

A node having some data to be transmitted, hold the token message and transmits the data, after completion of data transmission, the token is forwarded to next node. If a node does not have any data to send, it simply forwards the token to next node. Without holding the token data transmission is not allowed in ring topology.



Figure 9: Ring Topology

Some of the advantages of ring topology are:

- Minimum chances of data collision, as transmission is restricted with the help of token message.
- It is cheap to install.
-

Some of the disadvantages of ring topology are:

- Fault diagnosis is difficult in ring topology.
- Extension of the topology by adding a new device or removing an existing device is not easy.

Mesh Topology:

In a mesh topology each device/node is directly connected to every other device/node. It is one of the most reliable and fault tolerant network topology. To connect n device/nodes in a mesh topology, it requires $n(n-1)/2$ number of cables.



Figure 10: Mesh Topology

Some of the advantages of mesh topology are:

- Mesh topology is highly robust.
- Easy to diagnose the fault.
- As data is sent to destination directly without any intermediate node, hence data transmission is secure in mesh topology.

Some of the disadvantages of Mesh topology:

- Installation of mesh topology is most costlier.
- Maintenance is not easy.
- For a large number of devices/nodes it is very difficult to manage mesh network.

1.4 NETWORK TOPOLOGY

The computer networking involves both hardware and software components. Networking system also deals with embedded chip level programming and electric pulses. Overall the computer networking system is very complex to understand, implement and to extend. Further to make it easy to learn, extend and to provide interoperability the network models were developed with hierarchical approach with multiple layers. Each layer is independent of other layers and responsible for task assigned to it. A layer can communicate to layers just above it and just below it hence extension in any layer affects only the layers directly in contact with it. In general There are two network models defined in the networking: OSI model and TCP/IP model.

- - OSI Models: OSI is Open System Interconnection. It is also known as reference model for networking. OSI model is not exactly based on the working of the networking instead it was designed for understanding purpose of the networking. OSI model is seven layer model. It was proposed by ISO (International Organization for Standards) in 1984 after the TCP/IP model.

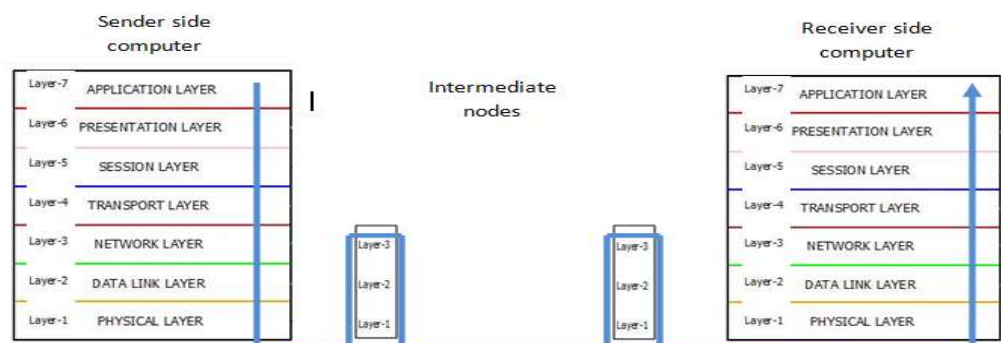


Figure 11: OSI reference model

- **Layer 1 - Physical Layer:** This layer is responsible for defining hardware related standards. Standards like cabling types, their

codings, connectors, ports etc are defined at this layer. Parameters related to physical transmission of data over medium i.e. signaling, pulse rate, bit rate, baud rate, power output etc. Layer 1 device is Hub.

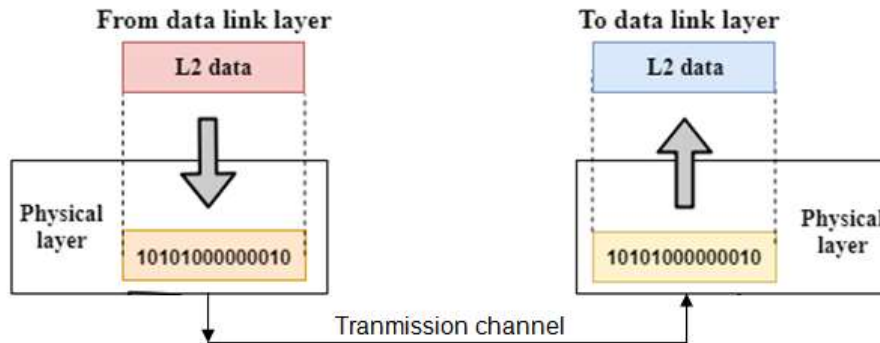


Figure 12 : Physical layer

Functions of a Physical layer:

- **Line Configuration:** how devices are physically connected to each other using wires or wireless channel is defined by line configuration.
- **Data Transmission:** transmission mode: simplex, or duplex (half or full) between two ends is defined by physical layer.
- **Topology:** physical arrangement of devices/ nodes defines the topology which is part of physical layer.
- **Signals:** Data is transmitted on physical medium in the form of electrical signals. The type of signals used to transmit the data is determined at physical layer.
- **Layer 2 – Data Link Layer (DLL):** This layer deals with both the hardware part and software part. This layer is mainly implemented in the Network Interface Card part of the computer. Data transmission between node to node is responsibility of this layer. It is also error control and flow control of traffic between two ends of the link. Frames are created specific to the underlying link. Physical address also known as the Media Access Control (MAC) address is used for recognizing nodes in the network for data delivery. MAC address is of 48 bits and is the property of the NIC. Out of 48 bits first 24 bits are recognizing the vendor created the NIC and rest 24 bits are unique for NICs of the vendor. Protocols used on this layer are: Ethernet, ATM etc. Layer 2 devices: Switch, Bridge etc.

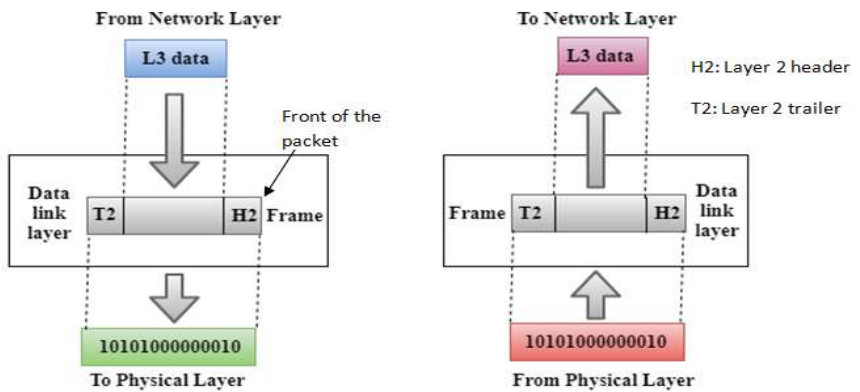


Figure 13: Data Link Layer

Functions of the Data-link layer

- **Framing:** DLL creates the physical medium specific frames by adding header and trailer to it. DLL is also responsible for creating frames from the physical bit stream received from physical layer on destination side.
- **Physical Addressing:** DLL includes physical address or MAC address of sender and receiver into header of the frame.
- **Flow Control:** DLL is responsible for flow control across the link.
- **Error Control:** DLL is also responsible for error control by adding a CRC (Cyclic Redundancy Check) value into the trailer part of the frame. Frames reported in error are retransmitted.
- **Access Control:** DLL allows transmission over a shared channel without collisions.

Layer 3 – Network Layer: Network layer is responsible for assigning IP addresses to networked devices. It is also responsible for best path selection from source to the destination for data delivery i.e. routing in the network. It provides connectionless services to the upper layers. Protocols used at this layer are: Internet Protocol (IP), IPv4 and IPv6, ICMP. Network layer devices are: Routers.

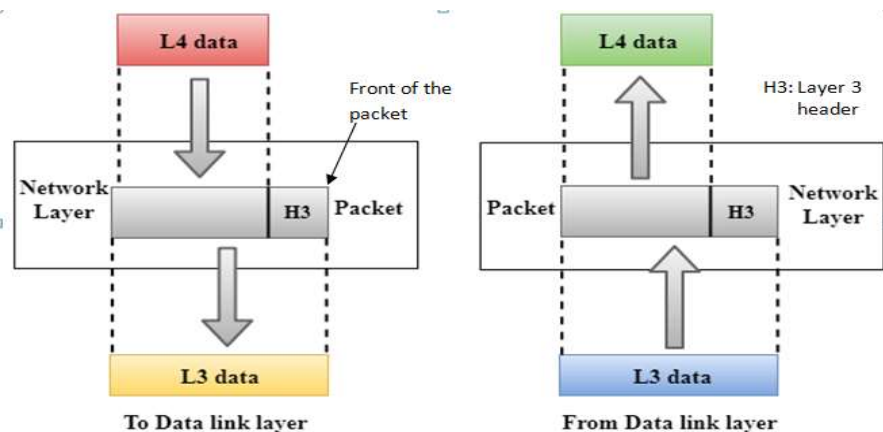


Figure 14: Network Layer

Functions of Network Layer:

- **Internetworking:** Network layer is responsible for data delivery between end hosts with the help of logical addressing i.e. IP addressing.
- **Addressing:** Network layer is responsible for assignment of the IP addresses to the networked devices. IP address is used to recognize a device in the Network.
- **Routing:** Routing is one of the major and important task of this layer. With the help of routing it determines the best optimal path between source and the destination.
- **Packetizing:** It receives the data from upper layer i.e. transport layer and adds its control information as header to form packets.

Layer 4 – Transport Layer: Transport layer is responsible for identification of the applications/ services by assigning the port numbers. End-to-end data delivery is the responsibility of the transport layer using sockets. A socket is combination of two entities: port number and IP address. This layer provides two kinds of data delivery services: connection oriented (reliable data delivery) and connectionless (unreliable data delivery). It is the 1st layer where user data is broken down into smaller segments. This layer receives formatted user data from upper layer i.e. session layer and adds its control information in the form of header to create segments. Protocols used at this layer are: **Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).**

TCP provides connection oriented reliable data delivery. It establishes a logical/virtual connection between end hosts before transmitting the data. TCP is also able to deliver segments in order (same as sent) and ensures the delivery of all the segments without error. Unlike TCP the UDP provides unreliable connectionless services. It does not guarantee the data delivery. It is the responsibility of the application to check the correctness of the data received also it is not responsible for ordered delivery of the segments. This protocol is used for the application which can tolerate the data loss. Data delivery using UDP is faster than TCP with less control information overhead. The header inserted by UDP is smaller than that of TCP.

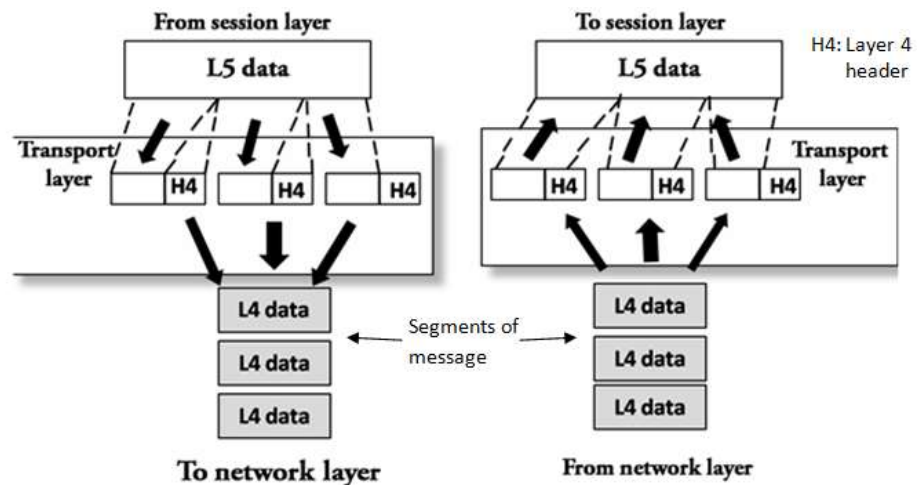


Figure 15: Transport Layer

Functions of Transport Layer:

- **Service-point addressing:** transport layer assigns a unique number: port number, to each computer networking application (even this port number is assigned to each of the tab of the web browser). This port number is added into header as control information. As discussed earlier the network layer is responsible to deliver the data to the computer (with the help of IP address) whereas the transport layer is responsible to collect the data from network layer and handover it to the correct application/process within the computer (at destination).
- **Segmentation and reassembly:** On the sender side transport layer divides the formatted user message into smaller segments, and assigns a sequence number that uniquely identifies each segment of the message. This sequence number is used to reassemble these segments on the destination side. Also this sequence number is used to confirm whether all the segments are delivered or not and are in order or out of order.
- **Connection oriented services:** Transport layer provides connection oriented services with the help of TCP. A logical end-to-end (application to application or socket to socket) connection is established before sending the user data. These services are used to provide reliable in order data delivery services.
- **Connectionless services: using UDP** transport layer can deliver data without bothering about the correctness and order of the segments received. It is the responsibility of the application layer to rearrange the segments to bring segments in order.
- **Flow control:** This layer provides flow control mechanism between end-to-end hosts. TCP windows are used to provide flow control facility.
- **Error control:** Unlike DLL which provides error control across the link, transport layer provides the error control facility between end-to-end hosts.

- **Layer 3 – Session Layer:** Session layer is responsible for maintaining sessions between hosts. It establishes, maintains and synchronizes the communication between end hosts. Ex. once user credentials are verified, the session layer at remote host keeps this session alive according to some predefined conditions and doesn't prompt for user credentials again until this session expires.

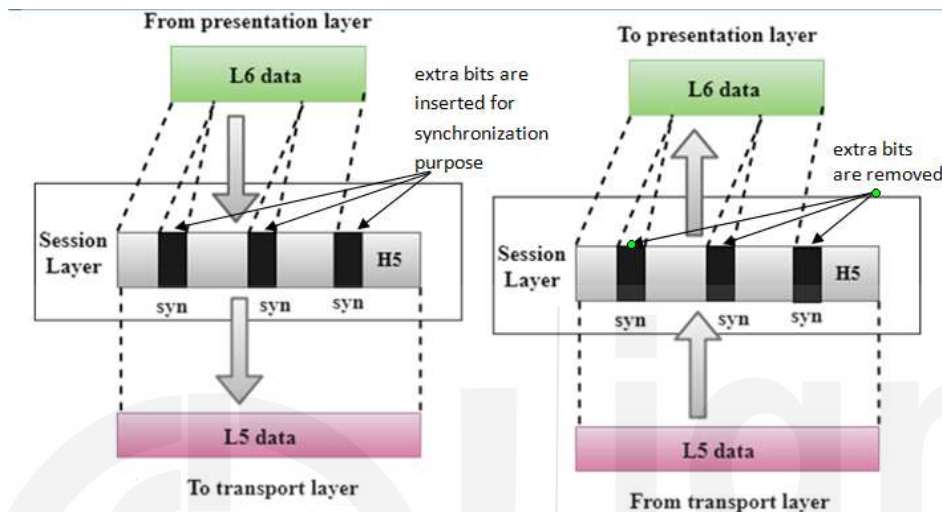


Figure 16: Session Layer

Functions of Session layer:

- **Dialog control:** Session layer provides two types of communication: half duplex and full duplex. It also records whose turn is it to transmit the data.
- **Synchronization:** Session layer is also responsible for synchronizing the data transmission by adding checkpoints while transmitting. If some occurs in between transmission, the system is rolled back to the checkpoint and retransmission begins from here again.
- **Layer 2 – Presentation Layer:** On the sender side this layer formats the message received from upper layer i.e. application layer and providing this formatted message to the underlying session layer. It defines the syntax and semantics of the message exchanged. On the destination side this layer receives the data from underlying layer i.e. session layer and reverses the operations of formatting to get the user raw message and provides it to the concerned application. Sometimes this layer is also called as the syntax layer.

Functions of Presentation layer:

- **Translation:** Session layer translates data from sender system's format to a generic format and reverse translating from the generic format to destination system's format.
- **Encryption:** Encryption and decryption of the message is the responsibility of presentation layer. Message encryption is performed on the sender side for security purpose and the message is decrypted on the destination side at presentation layer before handling it to the concerned application.
- **Compression:** At session layer data compression process takes place on the sender side and on the receiver side message will be decompressed based on the compression techniques used (lossy or lossless). Images are compressed with jpeg technique in the Internet.

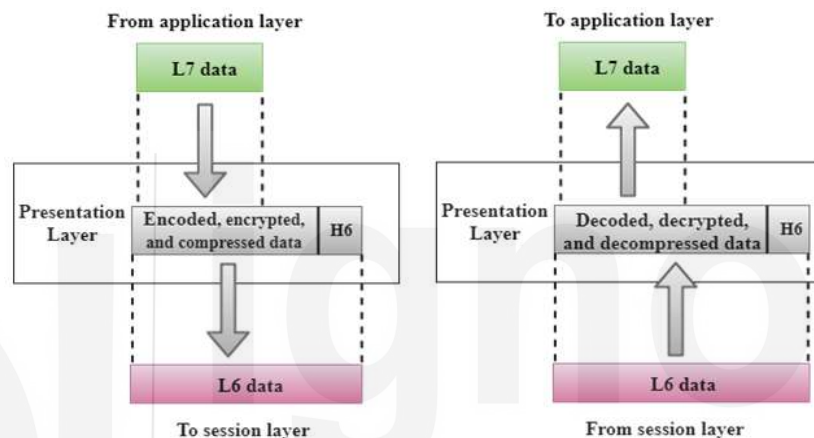


Figure 17: Presentation Layer

- Layer 1 – Application Layer: Application layer provides an interface to the user to interact and avail the Internet services. To be precise the application layer is not an application, rather it carry out the functionalities of this layer. Some of the application layer protocols are: HTTP, FTP, SMTP, Telnet etc.

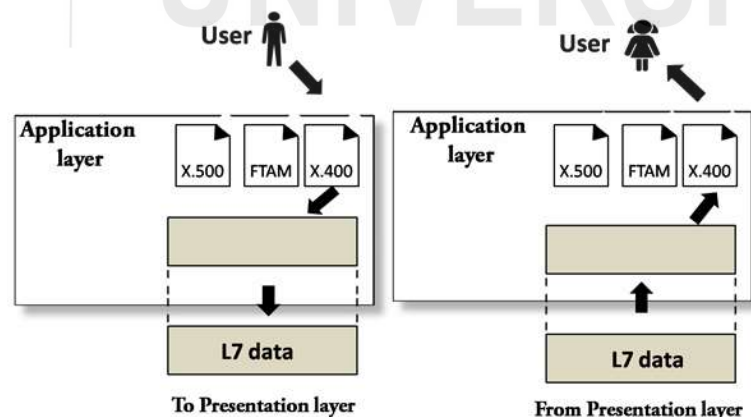


Figure 18: Application Layer

Functions of Application layer:

This layer provides performs user oriented services. Some of the commonly used application layer services are:

- **File transfer, access, and management (FTAM):** Application layer provides functionalities to download, upload and manage files places on a remote system through Internet.
- **Mail services:** Application layer is responsible to provides the facility of sending, receiving and storing emails in the Internet.

- TCP/IP Model

The TCP/IP networking model was introduced before the OSI reference model. The TCP/IP model was abstract in nature as compared to the OSI model. The TCP/IP model has four layers namely: application layer, transport layer, Internet layer and link layer. Layers of TCP/IP model are recognized by their names unlike as recognized by their layer number in OSI reference model. TCP/IP model is more closely defined to the actual working of the networking. Layers 7, 6 and 5 of OSI reference model are combined and named application layer in TCP/IP model. Rest of the layers are similar in both the models.

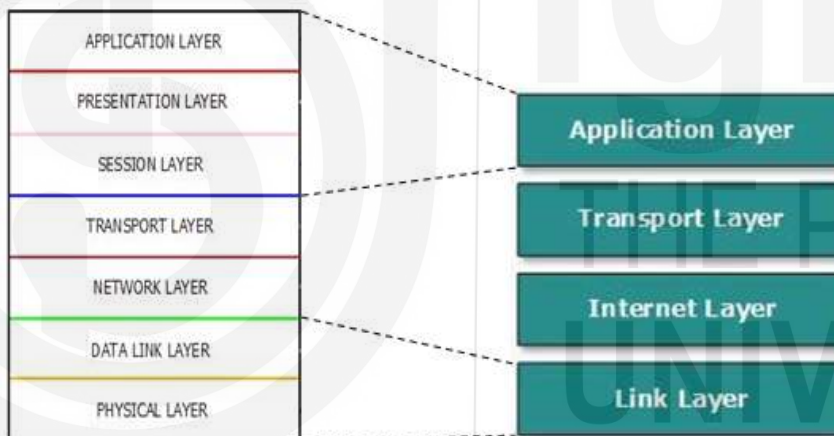


Figure 19: TCP/IP Networking Model

- **Application Layer:** It defines the interface for the user to access the Internet services. This layer is also responsible for formatting of the message to be transmitted and maintenance of the sessions. Ex. FTP, SMTP, HTTP etc.
- **Transport Layer:** Transport layer is responsible for process to process data delivery. s Transmission Control Protocol (TCP) and User datagram Protocol (UDP) are two major protocols of the transport layer.
- **Internet Layer:** Internet layer is responsible for host addressing and recognition in the Internet by assigning IP address. The routing is defined at this layer.

- **Link Layer:** Link layer is combination of DLL and Physical layer of OSI reference model.

Check your progress 2

Q.1 Explain World Wide Web (WWW) ?

Q.2 What is ring and star topology? What are their advantages?

Q.3 What are different functions of Data link Layer?

Q.4 Explain TCP/IP model with layer functionality.

Q.5 Access network connects to the Internet via tier -3 ISPs also known as?

- a) Connection ISPs
- b) Access ISPs
- c) Internetwork ISPs
- d) ISDNs

Q.6 Protocol used in wireless topologies?

- a) IEEE 802.11
- b) IEEE 802.12
- c) IEEE 802.13
- d) None

Q.7 A Wi-Fi enabled modem has a typical range of approx?

- a) 490 meters.
- b) 290 meters.
- c) 90 meters.
- d) 250 meters.

Q.8 Which of the following topology is also called a line or backbone topology?

- a) Ring
- b) Star
- c) Bus

d) Mesh

Q.9 Which layer is responsible for assigning IP addresses to networked devices?

a) Transport Layer

b) Data Link Layer

c) Physical Layer

d) Network Layer

1.5 SUMMARY

In this unit we have discussed about the Internet. Internet is network of networks. It is the world's largest network. Further, we have discussed about the public and private networks. Public networks are open to all i.e. anyone can access a public network. A private network is restricted in nature i.e. authentication parameters are required to get access of a private network. Internet access methods are also discussed in this unit. The Internet could be accessed through telephone lines, cable network and wireless network. Using telephone lines Internet can be accessed using dial-up setup, ISDN or DSL/ADSL setups. Network Topology is defined as the physical or logical arrangement of the devices in a network. Commonly four types of network topologies namely: Star topology, Bus topology, Ring topology and Mesh topology are discussed in this unit. The two network models namely: OSI model and TCP/IP model are discussed in this unit.

1.6 SOLUTIONS/ANSWERS

Answers to CYP 1:

Ans 1: Internet is a network of networks of broad area category. It is the largest network worldwide. It includes many LANs, WANs, MANs, and millions of computers. No single authority controls it; rather the local, national authority controls every segment of the Internet.

Ans 2: An Internet protocol suite is a set of standard protocols used for transmitting data over the Internet. Standard Internet protocols may be classified into four major categories:

1. **Application layer** – Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP).
2. **Transport layer** – Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
3. **Network layer** – Internet Protocol (IP) version 4 and version 6
4. **Data link layer** – Ethernet, IEEE 802.11 (Wi-Fi)

Ans 3: DSL is used for transmitting information at fast speed using the existing telephone lines. The user can use both telephone and computer data

transmission at the same time over the same line. A DSL modem is required to be installed at the end of the line to enable the computer to connect to Internet. Unlike dial-up line there is no need to dial the telephone number of the other end to establish the connection, rather this type of connection is always-on connection.

Asymmetric digital subscriber line (ADSL) is a type of DSL widely used for Internet connection. One of the reason of popularity of ADSL is the higher downstream (downloading) rate than the upstream (uploading) rate. Most of the Internet users use the Internet services to download the contents from Internet making it more popular.

Answers to CYP 2:

Ans 1: World Wide Web (WWW) provides a way to retrieve documents located on servers located on different geographical locations over the internet. WWW documents may include audio, text, visuals, graphics and links to other documents (hyperlinks). Hyperlinks are used to navigate among the documents.

Ans 2: In a ring topology, each device/computer is connected with two devices forming a ring type of structure. In general the data can move in one direction in the ring, but by installing another cable the data can move in reverse direction also. A token message keeps on rotating in the ring topology. A node having some data to be transmitted, hold the token message and transmits the data, after completion of data transmission, the token is forwarded to next node. If a node does not have any data to send, it simply forwards the token to next node. Without holding the token data transmission is not allowed in ring topology.

In a star topology, all the computers/devices are connected to a central switch/hub/server forming a star like topology. All the communication in star topology take place through the central node. The centralized node can act as a simple forwarding device or can also act as an intelligent firewall / filtering device.

Advantages of ring topology:

- Minimum chances of data collision, as transmission is restricted with the help of token message.
- It is cheap to install and configure.

Advantages of star topology:

- Easy to install and extend.
- Robust toward fault tolerance.

Ans 3: Data Link Layer perform following functions:

- **Framing:** DLL creates the physical medium specific frames by adding header and trailer to it. DLL is also responsible for creating frames from the physical bit stream received from physical layer on destination side.
- **Physical Addressing:** DLL includes physical address or MAC address of sender and receiver into header of the frame.
- **Flow Control:** DLL is responsible for flow control across the link.
- **Error Control:** DLL is also responsible for error control by adding a CRC (Cyclic Redundancy Check) value into the trailer part of the frame. Frames reported in error are retransmitted.
- **Access Control:** DLL allows transmission over a shared channel without collisions.

Ans 4: The TCP/IP networking model was introduced before the OSI reference model. The TCP/IP model was abstract in nature as compared to the OSI model. The TCP/IP model has four layers namely: application layer, transport layer, Internet layer and link layer. Layers of TCP/IP model are recognized by their names unlike as recognized by their layer number in OSI reference model. TCP/IP model is more closely defined to the actual working of the networking. Layers 7, 6 and 5 of OSI reference model are combined and named application layer in TCP/IP model. Rest of the layers are similar in both the models.

Layer Functionality:

- **Application Layer:** It defines the interface for the user to access the Internet services. This layer is also responsible for formatting of the message to be transmitted and maintenance of the sessions. Ex. FTP, SMTP, HTTP etc.
- **Transport Layer:** Transport layer is responsible for process to process data delivery. s Transmission Control Protocol (TCP) and User datagram Protocol (UDP) are two major protocols of the transport layer.
- **Internet Layer:** Internet layer is responsible for host addressing and recognition in the Internet by assigning IP address. The routing is defined at this layer.
- **Link Layer:** Link layer is combination of DLL and Physical layer of OSI reference model.

5. B, 6. A, 7. C, 8. C, 9. D

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